INCH POUND

MIL-R-70589B (AR)
30 September 1991
SUPERSEDING
(See 6.6)

MILITARY SPECIFICATION

REMOTE BIT TEST SET: TS-4211/TSM-100B: 9361217

This specification is approved for use by the U. S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements and tests for Remote Bit Tester, hereinafter referred to as the RBT, which forms a part of the Electronics System Test Set type AN/TSM-100B. The Remote Bit Tester is used to test all the BIT units that contain the 6402 UART. This tester is able to continuously transmit two digital bytes (one digital word) In succession with or without a time delay between words. A 500 KHz clock is produced on board with a TTL crystal oscillator. This is the same clock frequency used by all the UART BIT boards in the M167A2 and M163A2 systems (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-F-13926 - Fire Control Materiel, Manufacture and Inspection, General Specification for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4931
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:45Z
Check the source to verify that this is the current version before use.

MIL-I-45607 - Inspection Equipment, Acquisition, Maintenance and Disposition of
MIL-C-70594 - Component Board, Remote Bit Tester: 9361247

STANDARDS

MILITARY

MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-810 - Environmental Test Methods and Engineering Guidelines
MIL-STD-2000 - Standard Requirements for Soldering Electrical and Electronic Assemblies

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 41), 700 Robbins Avenue, Philadelphia, PA 19111–5094)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS (See 6.4)

U. S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

9361163 - Remote Bit Tester Schematic
9361217 - Remote Bit Tester Assembly
9361359 - RBT Test Connector Assembly

SPECIAL PACKAGING INSTRUCTION

PS9361217 - Packaging of Remote Bit Tester

(Copies of drawings, publications and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of conflict between the text of this document and the references cited herein the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:45Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawing 9361217, all associated drawings, and with all requirements specified in applicable specifications and standards. where applicable, the RBT shall meet the requirements of MIL-F-13926 as follows:

a. Dimensions and tolerances.
b. Inorganic protective surface finishings.
c. Part identification and marking.
d. Workmanship.

3.3 Environmental service conditions.

3.3.1 Vibration. The RBT shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to sinusoidal vibrations, in each of three mutually perpendicular axes, as follows:

Frequency Range	Amplitude
5 to 15 hertz	0.06 inches double amplitude
15 to 25 hertz	0.04 inches double amplitude
25 to 55 hertz	0.02 inches double amplitude

The frequency of vibration shall be swept logarithmically.
The cycling time shall be 120 minutes per axis.
The sweep rate shall be 3.75 ± 10% per decade.
The 5-55-5 hertz sweep time is 7 minutes 35 seconds ± 10%.

3.3.2 Shock. The RBT shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to a total of 18 impulse shocks consisting of three shock impulses of 15 g's ± 10%, 11 milliseconds ± 10%, half-sine wave applied in each direction of 3 mutually perpendicular axes.

3.3 Altitude.

3.3.3.1 Operating. The RBT shall meet the requirements of 3.4 and 3.5 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 Inches of Hg).

3.3.3.2 Non-operating. The RBT shall meet the requirements of 3.4 and 3.5 after exposure for not less than 10 minutes to barometric pressure encountered at 40,000 feet above sea level (5.5 inches of Hg).

Source: https://assist.dla.mil -- Downloaded: 202610-23T03:45Z
Check the source to verify that this is the current version before use.

3.3.4 Temperature .

3.3.4.1 Operating. The RBT shall meet the requirements of 3.4 and 3.5 when stabilized thermally at any temperature between +32°F and +131°F.

3.3.4.2 Non-operating (storage). The RBT shall not suffer permanent damage when subjected to the constant temperature extremes of -60°F for a period of 24 hours and +160°F for a period of 4 hours (after temperature stabilization) , and shall meet the requirements of 3.4 and 3.5 when returned to room ambient.

3.3.5 Humidity (non-operating). The RBT shall meet the requirements of 3.4 and 3.5 after exposure to an atmosphere with a relative humidity in excess of 95 percent for a period of 10 days,

3.4 Electrical. Each conductor shall be electrically continuous as specified on the RBT schematic diagram, 9361163.

3.5 Performance.

3.5.1 Functional description. The RBT shall have a quad differential to TTL level converter that has a 150 ohm resistor terminated at the inputs to match the line impedance and reduce ringing. The output shall be fed into the UART U3 Receiver Register Input. A quad TTL to differential level converter shall be used to send Transmitter Register Out (TRO) of the UART, and the 500 KHz system clock to the unit under test. The system clock shall be generated by a 500 KHz crystal oscillator. The output shall be buffered by two NAND gates which shall allow the user to inhibit the internal 500 KHZ clock and inject their own clock frequency. On power-up a pair of reset pulses shall be generated. A capacitor shall charge through a resistor and cause a Schmidt triggered inverter to go logic high to low when the threshold reaches from 1.4 to 1.9 v. This shall generate Reset (RST) and cause U11 pin 4 to go logic low to high and terminate Reset (RST). The UART is a CMOS device and requires pull-up resistors on all inputs to ensure that the threshold for a high is greater than 3 V. It is configured to transmit and receive eight data bits with odd parity. The transmissions and receptions have one start bit (low) and two stop bits (high). This format results from tying UART pins 34, 36, 37 and 38 to +5 V, and pins 4, 16, 35 and 39 to ground. Each bit from and to the UART (TRO & RRI) shall be 1/16 times the clock frequency. After the data is received serially it shall be converted to parallel form eight bits wide. This data shall appear on the Receiver Buffer Register (RBR). This data shall be inverted and shall go to the inputs of the Odd Parity Latch and the Even Parity Latch. Parity Error (PE) shall be evaluated; a high shall indicate even parity. A half-clock later, Data Ready (DR) shall go logic high. On the next rising clock edge after DR goes logic high, a flip flop shall toggle pin 6 low, generating Data Ready Reset (DRR). PR shall then fall low and the next rising clock edge

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:45Z
Check the source to verify that this is the current version before use.

will cause DRR to return high. The rising edge on pin 5 of the flip-flop shall also be accompanied by pin 6 rising. This rising edge, gated together with PE through two NAND gates shall generate either Odd Parity Latch Enable (OPLE) or Even Parity Latch Enable (EPLE). when PE is low, OPLE shall be generated; when PE is high, EPLE shall be generated. when switch S3 is placed in the NORMAL TEST MODE position, the data from the UART shall be fed through a tri-state buffer into digital-to-analog converters. When OPLE is low, a D/A converter shall convert the data on the bus into an analog voltage between 0 and 10 V. On the rising edge of OPLE this data shall be latched by the D/A converter. when EPLE is low, a D/A converter shall convert the data on the bus into an analog voltage between 0 and 10 V. On the rising edge of EPLE this data shall be latched by the converter. When S3 is placed in the UNIT 4 TEST MODE position, the buffer shall be put into tri-state. The outputs of the latches shall be sent to open collector buffers which drive individual LEDs through 270 ohm current limit resistors. A LED lighting shall be equivalent to having a logic high on the RBR pin of the UART. Switch S2 shall perform a dual function. It shall provide a manual reset to the circuit, and it shall clear the latches, causing all the LEDs to light (LAMP TEST). To send data to the unit under test, two bytes shall be loaded into the UART via four station hexadecimal encoded thumbwheel rotary switches S4. Stations 1 and 2 shall go through one tri-state buffer, while stations 3 and 4 shall go through another tri-state buffer and are tied together to the Transmitter Buffer Register (TBR) of the UART. When S2 is placed in the TEST position, pin 5 of a flip-flop shall go logic high; on the next rising clock edge pin 9 of this flip-flop (SEND) shall also go logic high. SEND and Transmitter Register Empty (TRE) shall be gated through an AND gate and the output is gated with DELAY through U20. This output shall be sent to the Johnson counter. The output of the Johnson counter shall be gated with the 500 KHz clock to create Transmitter Buffer Register Load (TBRL). TBRL shall be fed through an inverter to become TBRL, and is sent to the UART. TBRL shall also be sent to flip-flop U13; each rising edge will toggle the outputs, Buffer 1 Enable and Buffer 2 Enable. Buffer 1 Enable (B1E) when low shall enable buffer U19. Buffer 2 Enable (B2E) when low shall enable buffer U18. when switch S3 is placed in UNIT 4 TEST MODE, the transmitter delay circuitry shall be enabled. on the rising edge of B2E, pin 6 of the flip-flop (DELAY) will go logic low. DELAY shall allow counters U29 and U30 to count. when pin 9 of U30 goes logic high after 32.768 ms, pin 8 of U16 shall go logic low, which in turn will cause pin 11 of U31 to go logic low, this shall cause pin 6 of U16 (DELAY) to go logic high and enable the Johnson counter. when switch S3 is placed in the UNIT 4 TEST MODE position, only one D/A converter shall be updated through buffer U28 unless Byte 1 Data bit 6 (B1D6) from station 2 of the thumbwheel switch S4 is logic high. This latter condition shall allow both D/A converters to operate and receive data. The +5 v is provided by de-to-de switching regulator PS1. The +15 V is

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:45Z
Check the source to verify that this is the current version before use.

provided by PS2, which is a linear regulator. Inputs to both regulators run off the 115 V, 400 Hz from the voltage converter. Transient overvoltage spike protection is provided by transzorb D1.

3.5.2 UNIT UNDER TEST connector (J1). An 18 pin connector where the unit under test is connected to the RBT. The pin assignment shall be as shown in Table I (J1 connector), in accordance with schematic 9361163.

TABLE I. J1 connector.

PIN	FUNCTION
A	RCV
B	$\overline{\text{RCV}}$
D	DCLK
E	$\overline{\text{DCLK}}$
F	CLK 1
G	XMT
H	$\overline{\text{XMT}}$
K	$\overline{\text{EXT CLK}}$
L	$\overline{\text{INHIBIT OSC}}$
N	+5 V
P	+5 V RTN
R	CLK2
S	+5 V
T	+5 V RTN

3.5.3 Input/output power. with 115 VAC at either 60 Hz or 400 HZ applied to connector P5 pins C (HI), A (LO), and B (GND); the RBT shall operate and the lamp in the POWER ON/OFF switch (S5) shall illuminate when this pushbutton POWER switch (S5) is depressed to the ON position. This applies power to both the +5 vdc and +15 Vdc power supplies so that +5.0 +/-.1Vdc shall then be present at the UNIT UNDER TEST Connector (P1) between pins N and P, and between pins S and T.

Source: https://assist.dla.mil -- Downloaded: 202610-23T03:45Z
Check the source to verify that this is the current version before use.

3.5.4 RESET\LAMP TEST switch (S2). This momentary pushbutton switch shall provide a manual reset to the RBT when depressed. All channel 1 and 2 indicator lamps (DS1-DS16) shall illuminate when the RESET/LAMP TEST switch (S2) is held in the depressed position and all these lamps shall extinguish when this switch is released. (Channel 1 and 2 lamps indicate and identify digital signal code or data on the output pin of the UART.)

3.5.5 STATION 1234 switch (S4). These four hexadecimal thumbwheel switches shall load two digital bytes into the component board for transmission to the individual circuit card to be tested.

3.5.6 SEND switch (S1). This single pole, double throw toggle switch when set to the SEND (up) position shall enable the transmission of a digital signal to the individual circuit card to be tested.

3.5.7 Charnel 1 output voltage. The channel 1 output voltage when measured between test points J2(+) and J3(-) shall range from 0.00 +/- .08 Vdc to +9.96 +/-.10 Vdc.

3.5.8 Channel 2 output voltage. The channel 2 output voltage when measured between test points J5(+) and J4(-) shall range from 0.00 +/- .08 Vdc to +9.96 +/-.10 Vdc.

3.5.9 UNIT 4 TEST/NORMAL TEST switch (S3). This single pole, double throw toggle switch when set to the UNIT 4 TEST position shall enable a 16 bit word for transmission to the individual circuit cart to be tested and when set to the NORMAL TEST position shall enable an 8 bit word for transmission to the individual circuit card to be tested.

3.5.10 Component board. The remote bit tester shall contain a circuit board that contains the UART, TTL clock oscillator and A/D converters. The circuit board shall meet the requirements of MIL-C-70594.

3.6 First article When specified in the contract or purchase order, a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4. 3).

3.7 Soldering. All soldering shall be in accordance with MIL-STD-2000.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:45Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, tile contractor may use his own or any other facilities suits'ble for the performance of the inspection. requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of Sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplied submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 General provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Special sampling inspection (see 4.5).

4.2.1 Inspection conditions. Unless otherwise specified, all examinations and tests—shall be conducted at the following standard ambient conditions:

a.	Temperature	$+50^{\circ}$ to $+90^{\circ}$F
b.	Relative humidity	90% RH Max.
c.	Pressure	28 to 32 inches of Hg

4.3 First article inspection.

4.3.1 First article submission. The contractor shall submit a first article sample as designated by the contracting officer consisting of three (3) RBTs for inspection in accordance with 4.3.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:45Z
Check the source to verify that this is the current version before use.

4.3.2 Inspections to be performed. First article assemblies, components and test specimens may be subjected by the Government to any or all of the examinations and tests specified in Table II (Functional Test) and Table III (Special Sampling) herein and to any or all requirements of the applicable drawings.

4.3.3 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any assembly, component or test specimen to comply with any of the stated requirements.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term “inspection lot” is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

4.4.2 Examinations and tests.

4.4.2.1 Components, assemblies and subassemblies. All components, assemblies and subassemblies shall be inspected in accordance with the inspection provisions contained in this specification and the Quality Assurance Provisions (QAP) listed in the Technical Data Package (TDP). In the absence of QAPs, the applicable quality assurance provisions of MIL-F-13926 shall apply.

4.4.2.2 Final acceptance inspection. subsequent to first article approval, examinations and tests related to Section 3 herein shall be performed on a single defect [individual characteristic) basis in accordance with sampling plans specified in 4.4.2.3 herein. Examination and test for packaging and marking shall be in accordance with Section 5 herein. The tabulated classification of defects in 4.4.2.3 shall constitute the minimum inspection to be performed by the contractor after first article approval and prior to Government acceptance or rejection by item or lot.

4.4.2.3 Functional tests. The tests in Table II shall be performed on a 100% basis.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:45Z
Check the source to verify that this is the current version before use.

TABLE II. Functional tests (100 percent inspection).

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
Critical:	None	N/A
Major:		
101. Electrical	3.4	4.7.2
102. Performance	3.5	4.7.3
Minor:	None	N/A

4.5 Special sampling. One RBT out of the first 50 produced and one out of each 100 produced thereafter shall be selected at random. The sample shall have met the requirements and tests of Table II and shall then meet the tests in Table III. Unless otherwise specified in the contract, the items shall be returned to the lot upon successful completion of testing.

TABLE III. Special sampling.

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
Critical:	None	N/A
Major:		
103. Vibration	3.3.1	4.7.1.1
104. Shock	3.3.2	4.7.1.2
105. Altitude	3.3.3	4.7.1.3
106. Temperature	3.3.4	4.7.1.4
107. Humidity	3.3.5	4.7.1.5
Minor:	None	N/A

4.5.1 Failure of sample. Should any one item of a special sampling fail to meet any of the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections to all production since the last test as well as to current production has been accomplished in accordance with the provisions of the contract.

4.6 Inspection equipment.

4.6.1 Government-furnished inspection equipment. Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

Source: https://assist.dla.mil -- Downloaded: 202610-23T03:45Z
Check the source to verify that this is the current version before use.

4.6.2 Contractor-furnished inspection equipment.

4.6.2.1 Government design. Unless otherwise specified in the contract. all inspection equipment identified by drawing number in specifications or QAPs forming a part of the contract shall be supplied by the contractor in accordance with the design specified. The contractor may however, propose alternatives to Government-designs for approval in accordance with 6.3.

4.6.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the “Methods of Inspection specified in 4.7 of this specification and with the component inspection procedures specified in “Examination” and “Test Facilities” requirements of MIL-F-13926 whenever Government designs are not specified (see 4.6.2.1). Since tolerance of test equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment. Contractor inspection equipment designs shall be submitted for approval in accordance with 6.3.

4.6.2.3 Test equipment. In conjunction with 4.6.2.1 and 4.6.2.2, the test equipment listed in Table IV (Test Equipment list) shall be used-in-the performance of the applicable test methods and procedures as specified in 4.7.

TABLE IV l Test equipment list.

ITEM NUMBER		DESCRIPTION
1.	Digital multi meter (DMM)	Minimum requirements: 5 1/2 digits, Vdc accuracy: .010% + 3 counts at 20 Vdc range. Fluke model 8840A or equivalent.
2.	RBT Test Connector	RBT Test Connector Assembly 9361359, 18 pin connector used to link RCV with XMT and RCV with XMT
3*	Power supply, AC to AC	115 Vat, 60 Hz and 400 Hz capability, single phase, 250 VA minimum.

Source: https://assist.dla.mil -- Downloaded: 202610-23T03:45Z
Check the source to verify that this is the current version before use.

4.7 Methods of inspection.

4.7.1 Environmental. Environmental tests shall be conducted in accordance with the test conditions, general test performance guidance, and individual test methods of MIL-STD-810, except as specified herein. Test facilities, chambers and apparatus used in conducting the tests contained in this specification shall be capable of meeting the conditions required.

4.7.1.1 Vibration. The test shall be performed as specified in Method 514, Category 8, Procedure I, of MIL-STD-810 as modified by 3.3.1 herein. In Step 1 of Procedure I only steps 4 through 8 of paragraph II-2.1 apply. The RBT shall be attached to the vibration generator directly or with a fixture, and securely held by its normal means of attachment. After vibration, the RBT shall be tested as specified in 4.7.2 and 4.7.3 herein to determine compliance with 3.3.1.

4.7.1.2 Shock. This test shall be performed as specified in procedure I, Section II of Method 516 of MIL-STD-810 except that:

a. The post test functional check (Step 5 of Procedure I) need not be conducted until completion of all shock impulses.

b. The shock pulses shall be 15 g's ± 10% amplitude and 11 ms ± 10% duration.

c. The maximum g shall occur at 5.5 milliseconds ± 10%.

d. The shock measuring system shall have a flat frequency response within ± 10% from 20 to 2,000 hertz.

After the application of all the shock impulses, the RBT shall be tested as specified in 4.7.2 and 4.7.3 to determine compliance with 3.3.2.

4.7.1.3 Altitude (operating and non-operating). This test shall be performed as specified in Procedures II and III, Method 500 of MIL-STD-810. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.3.

4.7.1.4 Temperature.

4.7.1.4.1 High (operating and non-operating). This test shall be performed as specified in Procedures I and II, Method 501 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

4.7.1.4.2 Low (operating and non-operating). This test shall be performed as specified in Procedures I and II, Method 502 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:45Z
Check the source to verify that this is the current version before use.

4.7.1.5 Humidity (non-operating). This test shall be performed in accordance with Procedure III, Method 507 of MIL-STD-810 except that in Step 7, checkout shall be performed at the conclusion of the test only, with the item at standard ambient conditions. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.5.

4.7.2 Electrical. Continuity shall be inspected by standard measuring equipment and visual examination for compliance with 3.4.

4.7.3 Performance.

4.7.3.1 Initial conditions. Set the POWER ON/OFF switch (S5) to the OFF position. Set the STATION 1234 switch (S4) to 0 0 0 0. Set the SEND switch (S1) to the SEND (up) position. Set the UNIT 4 TEST/NORMAL TEST switch (S3) to the NORMAL TEST position. Connect power supply (item 3) to the RBT connector P5 pins C (HI), A (LO), and B (GND).

4.7.3.2 Input/output power.

4.7.3.2.1 Input at 60 Hz. Setup the power supply (item 3) to apply 115 VAC at 60 Hz to the RBT. Energize the RBT by depressing the POWER ON/OFF switch (S5) to the ON position. verify that the RBT operates and that the lamp in the POWER switch (S5) illuminates. Using the DMM (item 1) verify that +5.0 +/-.1 Vdc is present at the UNIT UNDER TEST Connector (Pi) between pins N (+) and P (–), and between pins S (+) and T (-). Turn the RBT off by pushing the POWER switch (S5) to the OFF position.

4.7.3.2.2 Input at 400 Hz. Repeat the procedures in paragraph 4.7.3.2.1 with the power supply (item 3) setup to apply 115 VAC at 400 Hz to the RBT. After these tests, with the RBT off, attach the RBT Test Connector (item 2) to the UNIT UNDER TEST connector (Pi) on the RBT. Energize the RBT by depressing the POWER switch (S5) to the ON position and complete the performance testing with the input power at 400 Hz.

4.7.3.3 RESET/LAMP TEST switch (S2). With the RBT switch settings unchanged, verify that all the channel 1 and 2 indicator lamps (DSI-DS16) illuminate when the RESET/LAMP TEST switch (S2) is pushed in and held at the depressed position and verify that all these lamps extinguish when this switch is released.

4.7.3.4 STATION 1234 switch (S4). Without changing the RBT switch settings, set the STATION 1234 switch (S4) to 15 15 0 0 and verify that all the channel 1 and 2 indicator lamps (DS1-DS16) illuminate. Set the STATION 1234 switch (S4) to 0 0 0 0 and verify that all these lamps extinguish. Set the STATION 1234 switch (S4) to 0 0 15 15 and verify that all these lamps illuminate. Set the STATION 1234 switch (S4) back to 0 0 0 0 and verify that all these lamps extinguish.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:45Z
Check the source to verify that this is the current version before use.

4.7.3.5 SEND switch (S1). Without changing the other RBT switches, set the SEND switch (S1) to the down position then set the STATION 1234 switch (S4) to 15 15 15 15. Verify that all the channel 1 and 2 indicator lamps (DS1-DS16) remain extinguished until the SEND switch (S1) is set to the SEND (up) position and then verify that these lamps all illuminate.

4.7.3.6 Channel 1 output voltaae. with the STATION 1234 switch (S4) still set to 15 15 15 15 and the other RBT switches unchanged, connect the DMM (item 1) to test points J2(+) and J3(-). verify that the Channel 1 output voltage is +9.96 +/–.10 Vdc. Set the STATION 1234 switch (S4) to 0 0 0 0 and verify that the output voltage is 0.00 +/–.08 Vdc.

4.7.3.7 Channel 2 output voltage. Without changing the other RBT switches, set the STATION 1234 switch to 0 0 0 0, connect the DMM (item 1) to test points J5(+) and J4(-). Verify that the Channel 2 output voltage is +9.96 +/-.10 Vdc. Set the STATION 1234 switch (S4) to 0 0 O 0 and verify that the output voltage is 0.00 +/–. 08 Vdc.

4.7.3.8 UNIT 4 TEST/NORMAL TEST switch (S3). with the STATION 1234 switch (S4) still set to 0 0 0 0, set the UNIT 4 TEST/NORMAL TEST switch (S3) to the UNIT 4 TEST position. Set the STATION 1234 switch (S4) to 15 15 15 15 and verify that all the channel 1 and 2 indicator lamps (DS1-DS16) remain extinguished.

4.7.4 Component board. The component board shall have demonstrated compliance with MIL-C-70594.

4.8 Soldering. Soldering processes shall be inspected to the requirements and acceptance criteria of MIL-STD-2000 as specified in 3.7.

5. PACKAGING

5.1 General. Preservation, packing and marking shall be in accordance with packaging Data Sheet Instruction PS9361217. The level of protection shall be as specified in the procurement document.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The RBT is designed for slide-mounting into the AN\TSM-100B Electronics System Test Set, which is used for troubleshooting and performance checking of specific weapon system components. The remote BIT tester will be used to check out remote BIT circuit cards used on the M163A2 or M167A2 VADS.

Source: https://assist.dla.mil -- Downloaded: 2026l0-23T03:45Z
Check the source to verify that this is the currentversion before use.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).
c. Applicable stock number.
d. First article sample requirements.
e. Packaging requirements, if other than specified in Section 5.
f. Serialization requirements, if applicable.
g. Certificate of conformance for each lot or shipment of product.

6.3 Submission of contractor inspection equipment designs for approval. Submit two copies of designs as required to: Commander, U.S. Army AMCCOM, Attn: AMSMC-QAF-I, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research, Development and Engineering Center (ARDEC), may also Include drawings prepared by, and identified as, U.S. Army Armament Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.

6.5 Key word listing.

Fire Control System
Vulcan Air Defense System (VADS)
Self-Propelled M163A2, 20MM, VADS
Towed M167A2, 20MM, VADS

6.6 Changes from previous issue. The previous issues were for ARDEC internal use only. The were not forwarded to Naval publication and Forms center (NPFC) for publication. This issue is initial public release for unlimited distribution.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 4931-A292)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:45Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:45Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

1 I RECOMMEND A CHANGE:	[illegible]	2 DOCUMENT DATE (YYMMDD) 3[illegible] SEPTEMBER 1991

3 DOCUMENT TITLE

REMOTE BIT TEST SET: TS-4211/TSM-100B: 9361217

4 NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)

5 REASON FOR RECOMMENDATION

6 SUBMITTER

a NAME (Last, First, Middle Initial)	b ORGANIZATION	
c ADDRESS (Include Zip Code)	d TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7 DATE SUBMITTED (YYMMDD)

8 PREPARING ACTIVITY

a NAME US ARMY ARDEC STANDARDIZATION OFFICE	b TELEPHONE (Include Area Code) (1) Commercial: (201) 724-6674 (2) AUTOVON: 880-6674
ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S PICATINNY ARSENAL,NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

Source: https://assist.dla.mil -- Downloaded: 202640-23T03:45Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:45Z
Check the source to verify that this is the current version before use.